

AI Portfolio Copilot – Problem Framing Document (Phase 1)

1. Project Title

AI Portfolio Copilot: A Safety-First Decision Support Agent for Wealth Management

2. Industry Scenario Selected

Scenario: Business Operations — AI Operations Copilot (Decision Support Only)

This project aligns with a decision-support system where the AI agent assists wealth managers in analyzing portfolio data and preparing advisory insights. The agent does not execute transactions or modify any financial data and strictly operates within advisory boundaries, complying with safety-first requirements.

3. Primary User Persona

User Type

Certified Financial Planner (CFP) / Wealth Manager / Investment Advisor

User Profile

A financial professional responsible for managing client portfolios across equity, debt, and hybrid instruments. The user regularly conducts portfolio reviews, communicates with clients, and ensures that investments align with financial goals and risk profiles.

Daily Workflow

- Review portfolio performance and asset allocation
- Compare returns against benchmarks (Nifty, category averages, etc.)
- Identify concentration risks and allocation drift
- Prepare client review notes and presentations
- Answer client queries during volatile market conditions
- Suggest rebalancing strategies (subject to client approval)

Pain Points

- Time-consuming manual analysis of portfolio data
- Repetitive explanation of performance to clients
- Difficulty maintaining consistency in advisory communication

- Managing multiple client contexts simultaneously
- Handling uncertainty during volatile market periods

4. Problem Statement

Wealth managers spend significant time interpreting portfolio data, market trends, and client objectives to deliver consistent and explainable advisory insights. This process is repetitive, prone to inconsistency, and becomes challenging during volatile market conditions.

The objective of this project is to build an AI Portfolio Copilot that assists advisors by:

- Summarizing portfolio performance
- Explaining reasons for underperformance or outperformance
- Identifying allocation imbalances and concentration risks
- Answering client-style queries
- Suggesting discussion points for advisory meetings
- Escalating uncertain or high-risk scenarios

The agent operates strictly as a decision-support system, ensuring compliance with safety and regulatory expectations.

5. Workflow Supported

Workflow Name

Portfolio Review & Advisory Preparation Workflow

Process Flow

1. Advisor inputs portfolio data and client context
2. User queries the agent regarding performance, allocation, or risks
3. Agent retrieves relevant portfolio and knowledge data
4. Agent uses tools (calculations, checks) for analysis
5. Agent generates structured advisory insights
6. Agent maintains short-term context (client profile, goals)
7. Agent escalates uncertain or high-risk scenarios

6. Inputs, Outputs, Constraints, and Assumptions

6.1. Inputs

- Portfolio statements (holdings, allocation)
- Benchmark data
- Market outlook notes
- Client profile (risk, horizon, goals)
- Advisor queries in natural language

Input Type	Source
Portfolio Data	Uploaded PDF / Excel / Manual input
Benchmark Data	Preloaded dataset or documents
Market Notes	Research reports / outlook PDFs
Client Profile	Advisor input
Queries	User interaction (chat)

Example Input (Structured):

```
{  
  "client": "Client_A", "portfolio_value": 5000000,  
  "allocation": {"equity": 70, "debt": 20, "cash": 10}  
}
```

Note : All sensitive client identifiers are masked or anonymized before processing to ensure compliance with privacy and security requirements.

6.2. Outputs

- Portfolio summary
- Performance explanation
- Allocation and risk analysis
- Benchmark comparison
- Advisory discussion points
- Uncertainty or escalation statements

6.3. Constraints

The agent must:

- Not execute trades or financial transactions
- Not modify portfolio or client data
- Not provide guaranteed returns
- Not fabricate financial data
- Not provide legal or tax advice
- Not store personally identifiable information (PII) in logs
- Escalate ambiguous or high-risk queries

Note : Sensitive client data (PII) must not be stored, exposed, or logged without masking

Assumptions

- Portfolio data is available in structured or semi-structured format
- Benchmark data is accessible
- Advisor remains final decision-maker
- Initial version focuses on mutual fund and multi-asset portfolios

7. Example User Questions

1. Why did this portfolio underperform the benchmark this quarter?
2. Is the current allocation suitable for a moderate-risk investor?
3. Which holdings are contributing most to concentration risk?
4. What should I discuss with the client in the next review meeting?
5. Explain this portfolio performance in simple client-friendly language

8. Success Criteria

Functional Success

- Accurate portfolio interpretation
- Relevant information retrieval
- Correct use of analytical tools

- Consistent and structured responses
- Safe refusal of restricted actions

User Success

- Reduced time in preparing portfolio reviews
- Improved clarity in client communication
- Better identification of risks and allocation issues
- Consistency in advisory explanations

Technical Success

- Improvement from baseline to LLM-enabled responses
- Better accuracy with retrieval vs without retrieval
- Correct tool usage for calculations
- Effective memory usage in multi-turn interactions
- Demonstrated learning through feedback adaptation

9. Measurable Metrics

Metric	Description
Retrieval Accuracy	% of responses using relevant portfolio data
Response Correctness	% of factually accurate responses
Safety Compliance	% of unsafe queries correctly refused
Tool Usage Accuracy	% of correct tool selections
Context Retention	% of correct multi-turn responses
Adaptation Score	Improvement after feedback

10. Failure Cases and Edge Scenarios

Missing Data

- Risk: Incorrect assumptions
- Handling: Explicit uncertainty statement

Unsafe Request

- Example: "Execute portfolio rebalancing"
- Handling: Refusal with explanation

Ambiguous Query

- Risk: Misleading advice
- Handling: Escalation to advisor

Retrieval Failure

- Risk: Weak response
- Handling: State lack of sufficient data

Incorrect Tool Usage

- Risk: Wrong analysis
- Handling: Logging and correction

Memory Leakage

- Risk: Privacy issues
- Handling: Reset rules

Overconfidence

- Risk: Misleading statements
- Handling: Cautious language enforcement

PII Exposure

- Risk: Compliance violation
- Handling: Masking sensitive data

11. Safety Design Principles

The system is designed with a safety-first approach:

- Decision support only (no execution capability)
- No hallucination of financial data
- Clear uncertainty communication
- Mandatory escalation for complex scenarios

- Strict PII protection in logs
- No guaranteed return statements

12. Conclusion

The AI Portfolio Copilot is designed to enhance the efficiency and consistency of wealth management workflows by providing structured, explainable, and safe decision-support insights. The system focuses on augmenting human advisors rather than replacing them, ensuring that all critical financial decisions remain under human supervision.